VoltEdge

Stop guessing why energy spikes. Start knowing.

Enterprise energy intelligence for multi-site industrial and commercial operations.

Real-time Monitoring · Causal AI · CSRD-ready ESG · Anomaly Detection · Demand Forecasting

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01 THE PROBLEM WE SOLVE

Multi-site businesses face a critical gap: energy anomalies are flagged, but nobody can explain *why* they happened. CSRD compliance requires granular, auditable energy and emissions data — yet most operations still compile it manually, consuming weeks of analyst time every reporting cycle. VoltEdge closes that gap.

4	90	903	34,560	CSRD
Demo sites across 3 continents	Days of historical energy data	Automated tests passing	Hourly readings at launch	Compliance-ready ESG output

02 KEY FEATURES

Real-time site monitoring — Hourly kWh, voltage, current, power factor across all sites

Causal AI root-cause analysis — PC-algorithm causal graphs with quantified strength scores explain anomalies, not just flag them

Automated ESG reporting — GRI 302-1, Scope 2 location and market-based, carbon intensity — CSRD-ready

AI assistant — Natural language energy queries powered by Anthropic Claude

Demand forecasting — 24h consumption forecasts with 95% confidence intervals

Portfolio view — Cross-site comparison, heatmaps, anomaly benchmarking, demand response revenue estimates

Data export — CSV download of consumption and emissions data for auditors and regulators

Role-based access control — JWT auth with SOC 2 audit trail for enterprise deployments

03 TECHNICAL ARCHITECTURE

Dashboard	Plotly Dash 4 · Bootstrap 5 · Bootstrap Icons	**AI Assistant**	Anthropic Claude · Direct mode · No server required
API Layer	FastAPI · Uvicorn · Pydantic	**Data Layer**	SQLite (demo) · PostgreSQL-ready · 90-day history
ML — Anomaly	Isolation Forest · In-memory cache · 30-min TTL	**Auth & Audit**	JWT · RBAC · SOC 2 audit trail
ML — Forecasting	Gradient Boosting (scikit-learn GBM)	**Ingestion**	MQTT · CSV import · REST API
Causal AI	PC algorithm (NetworkX) · Root-cause analyser	**Deployment**	Docker · Render · Railway · GitHub CI/CD

04 MARKET OPPORTUNITY

Mid-size UK manufacturers — £1M–£50M annual energy spend, CSRD mandates

Data centre operators — 24/7 baseload, PUE monitoring, carbon intensity reporting

Logistics and warehouse operators — Multi-site, high energy intensity, extended hours

Commercial property portfolios — Hotels, retail chains, student accommodation

Energy consultancies (OEM) — White-label platform for 50–200 industrial clients

ESG and sustainability leads — Board-level CSRD accountability, audit-ready output

05 COMMERCIAL MODEL

Direct	OEM / White-label	Enterprise	API Licence
£1,500–£3,000	**£400–£600**	**£60k–£200k**	**Custom**
Per month · all sites	Per site / month	Per year · custom SLA	Per-call or flat rate
Mid-size operators	Energy consultancies	Large industrial groups	BMS / HR / ERP integrations

06 FUTURE ROADMAP

Live meter integration — Direct MQTT/API to AMR and smart meters

Carbon credit marketplace — Scope 3 tracking with verified carbon registries

Demand response automation — Automated DR dispatch linked to grid signals

Multi-tenant SaaS — Full tenant isolation and branded portals for OEM scale

Regulatory expansion — SEC climate rules, EU Taxonomy, ISO 50001 audit pack

Mobile application — Native iOS/Android for on-site energy managers

Live Demo	GitHub	Portfolio	LinkedIn
voltedge-1k6q.onrender.com	github.com/ShazMoghaddam/voltedge	shazmoghaddam.github.io	linkedin.com/in/shazmoghaddam